

```

if not self.aliens:
    # Destroy existing bullets and create new fleet.
    self.bullets.empty()
    self._create_fleet()

```

We've created a new method, `_check_bullet_alien_collisions()`, to look for collisions between bullets and aliens and to respond appropriately if the entire fleet has been destroyed. Doing so keeps `_update_bullets()` from growing too long and simplifies further development.

TRY IT YOURSELF

13-5. Sideways Shooter Part 2: We've come a long way since Exercise 12-6, *Sideways Shooter*. For this exercise, try to develop *Sideways Shooter* to the same point we've brought *Alien Invasion* to. Add a fleet of aliens, and make them move sideways toward the ship. Or, write code that places aliens at random positions along the right side of the screen and then sends them toward the ship. Also, write code that makes the aliens disappear when they're hit.

Ending the Game

What's the fun and challenge in playing a game you can't lose? If the player doesn't shoot down the fleet quickly enough, we'll have the aliens destroy the ship when they make contact. At the same time, we'll limit the number of ships a player can use, and we'll destroy the ship when an alien reaches the bottom of the screen. The game will end when the player has used up all their ships.

Detecting Alien-Ship Collisions

We'll start by checking for collisions between aliens and the ship so we can respond appropriately when an alien hits it. We'll check for alien-ship collisions immediately after updating the position of each alien in *Alien Invasion*:

```

alien
_invasion.py
def _update.aliens(self):
    --snip--
    self.aliens.update()

    # Look for alien-ship collisions.
    ❶ if pygame.sprite.spritecollideany(self.ship, self.aliens):
    ❷     print("Ship hit!!!")

```

The `spritecollideany()` function takes two arguments: a sprite and a group. The function looks for any member of the group that has collided with the sprite and stops looping through the group as soon as it finds one member that has collided with the sprite. Here, it loops through the group aliens and returns the first alien it finds that has collided with ship.